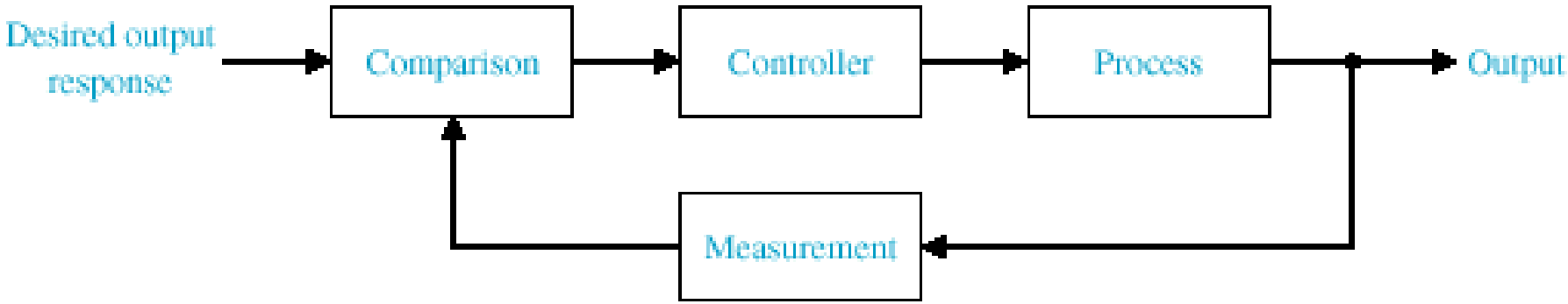
Hanani Abdul Wahab

BDA3073

Review

Control system is a group of components which maintains desired results (goals) by manipulating the value of another variable in the system.



Goals (Desired results)

- Stability: system maintains desired operating point
- Performance: system responds rapidly to changes
- Robustness: system tolerates perturbations in dynamics

Contents

- Introduction
- Dynamics of Mechanical Systems
- Dynamics of Electrical System
- Dynamics of Electromechanical System

Introduction

We should know the characteristic or behavior (input-output characteristics) of the controlled object (plant or system to be controlled) so that we can control the plant successfully!

In other word, we need a model of the plant so that we can design an appropriate controller.

What is model?

Model is collection of mathematical equation which represent behavior/characteristic of real system (physical system).

Introduction

- The term mathematical model, in the control engineering perspective, implies a set of differential equations that describe the dynamic behavior of a process.

How to get model of a system?

- The set of differential equations that describe the behavior of physical systems are typically obtained by utilizing the physical laws of the process.

Physical laws of the process $\rightarrow$ Differential equations

Introduction

What physical law do you need?

The physical law used depends on the system to be modeled.

Mechanical system $\rightarrow$ Newton's laws

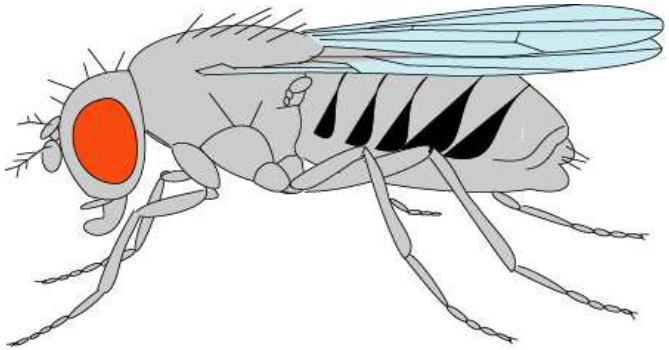
Electrical system $\rightarrow$ Kirchhoff's laws

Dynamics of Mechanical Systems

What is mechanical system?









Mechanical system is a system subjected by forces which cause motion.

Governed by Newton's Law

Dynamics of Mechanical Systems

Mechanical systems are concerned with the behavior of matter under the action of forces/torques.

Newton's laws are used to model the Mechanical system.

Translational Motion

$$\sum_{i=1}^{N} F_i = m\frac{d^2x}{dt^2}$$

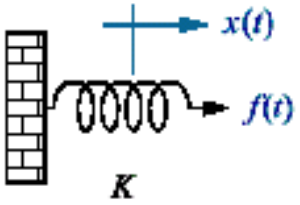
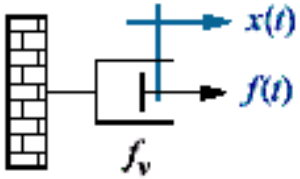
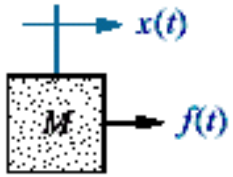
Rotational Motion

$$\sum_{i=1}^{N} T_i = J\frac{d^2\theta}{dt^2}$$

Dynamics of Mechanical Systems

Elements of Mechanical System

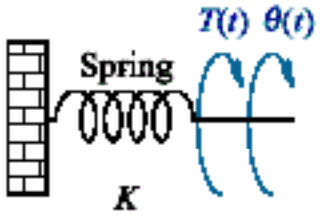
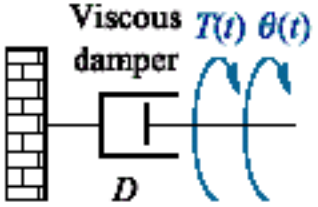
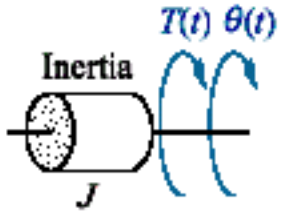
Translational systems

Component	Force-velocity	Force-displacement
Spring 	$f(t) = K\int_0^t v(\tau)\,d\tau$	$f(t) = Kx(t)$
Viscous damper 	$f(t) = f_v v(t)$	$f(t) = f_v \frac{dx(t)}{dt}$
Mass 	$f(t) = M\frac{dv(t)}{dt}$	$f(t) = M\frac{d^2x(t)}{dt^2}$

Dynamics of Mechanical Systems

Rotational Mechanical System

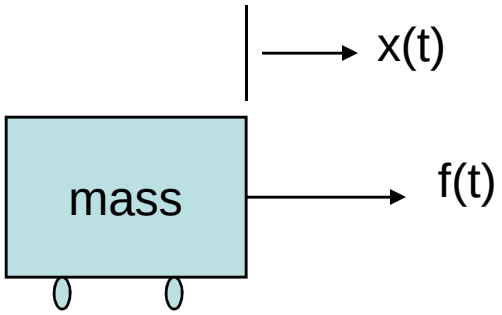
Rotational systems

Component	Torque-angular velocity	Torque-angular displacement
	$T(t) = K\int_0^t \omega(\tau)\,d\tau$	$T(t) = K\theta(t)$
	$T(t) = D\omega(t)$	$T(t) = D\frac{d\theta(t)}{dt}$
	$T(t) = J\frac{d\omega(t)}{dt}$	$T(t) = J\frac{d^2\theta(t)}{dt^2}$

Dynamics of Mechanical Systems

Translational Mechanical System

a) Mass

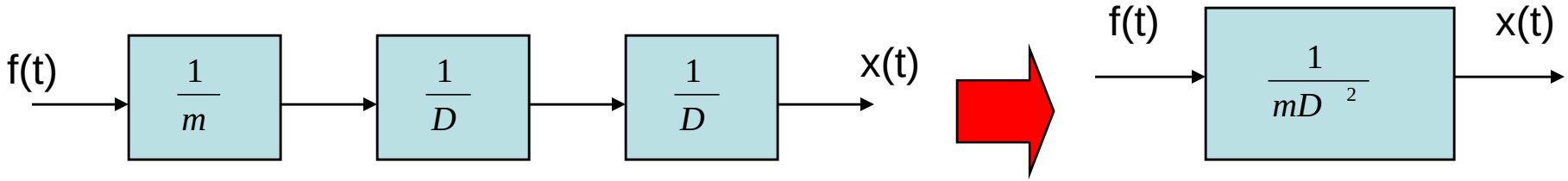


$$f(t) = m\frac{d^2x}{dt^2} \ldots\ldots displacement$$

$$f(t) = m\frac{dv}{dt} \ldots\ldots velocity$$

$$f(t) = ma \ldots\ldots\ldots accelaration$$

Block diagram $D = d/dt = \frac{1}{s}$



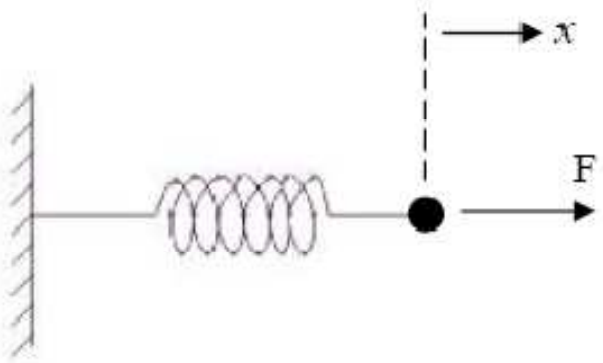
Dynamics of Mechanical Systems

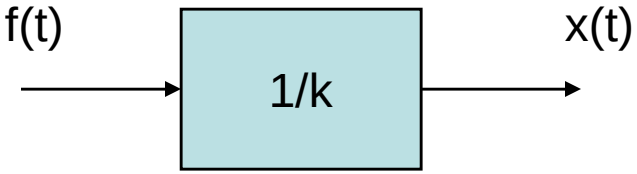
Translational Mechanical System

b) Spring

$$f(t) = kx(t).....displacement$$

$$f(t) = k\int vdt.....velocity$$





Block Diagram

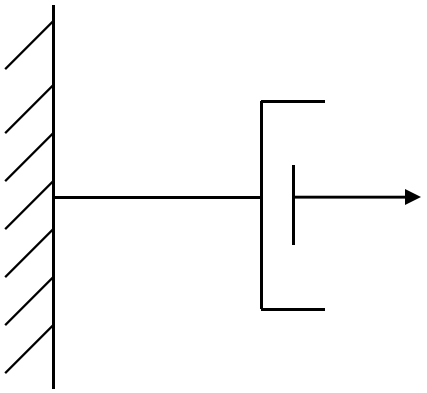
Dynamics of Mechanical Systems

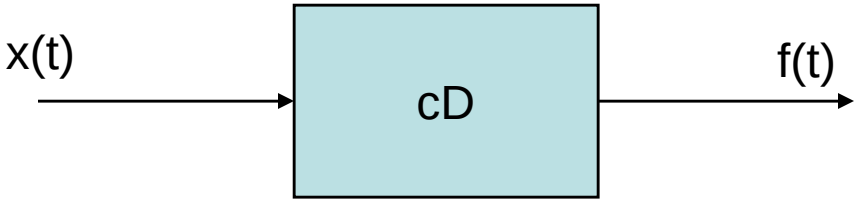
Translational Mechanical System

c) Viscous Damper

$$f(t) = c\frac{dx}{dt} \text{.....} displacement$$

$$f(t) = cv(t) \text{.....} velocity$$





Block Diagram

Dynamics of Mechanical Systems

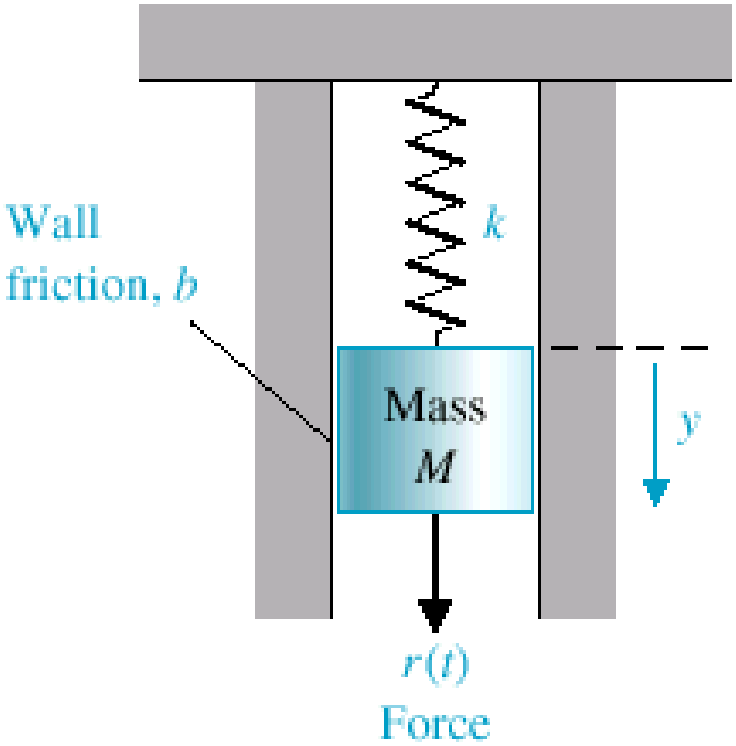
Modeling Process

- Assign variable such as x and θ that are both necessary and sufficient to describe an arbitrary position of the system.
- Draw free body diagram of each components. Indicate all forces acting on each body and their reference directions. Also indicate the accelerations of the center of mass.
- Apply Newton's law for translation or rotation.
- Combine the equations to eliminate internal forces.
- The number of independent equation should equal to the number of unknown.

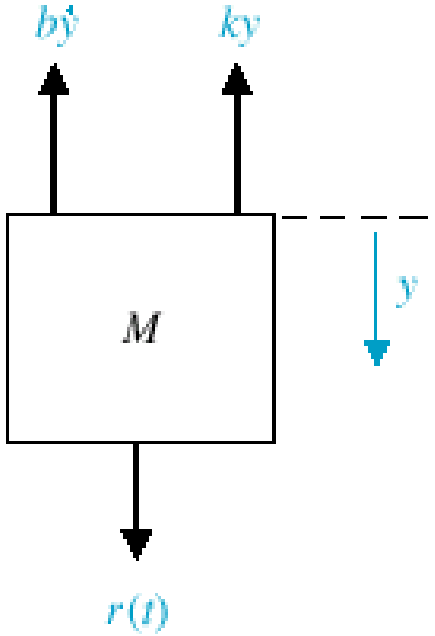
Dynamics of Mechanical Systems

Example 1

Obtain the mathematical model!



FBD?

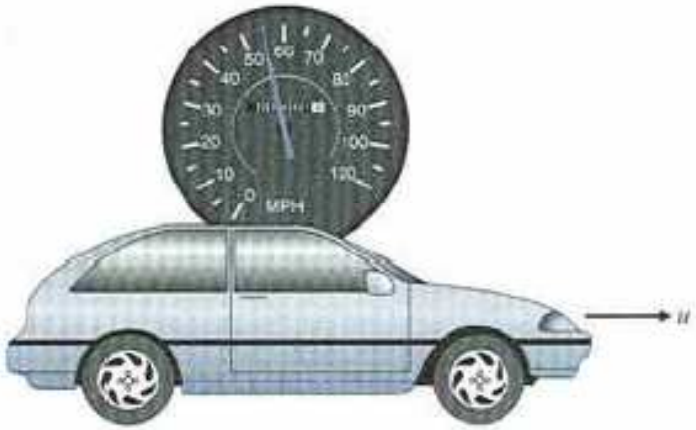


Assumption: wall friction is a viscous force (damper)

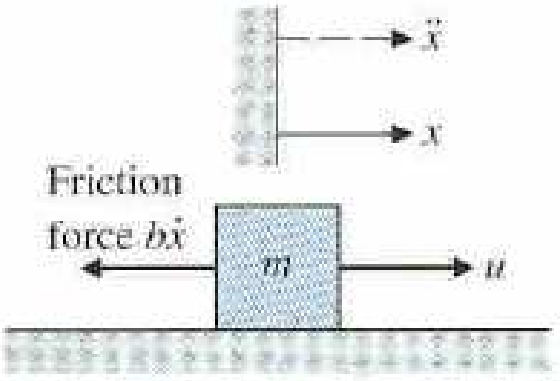
Dynamics of Mechanical Systems

Example 2

Obtain the mathematical model for the speed and forward motion of the car!



FBD?



In term of position

$$\ddot{x} + \frac{b}{m}\dot{x} = \frac{u}{m}$$

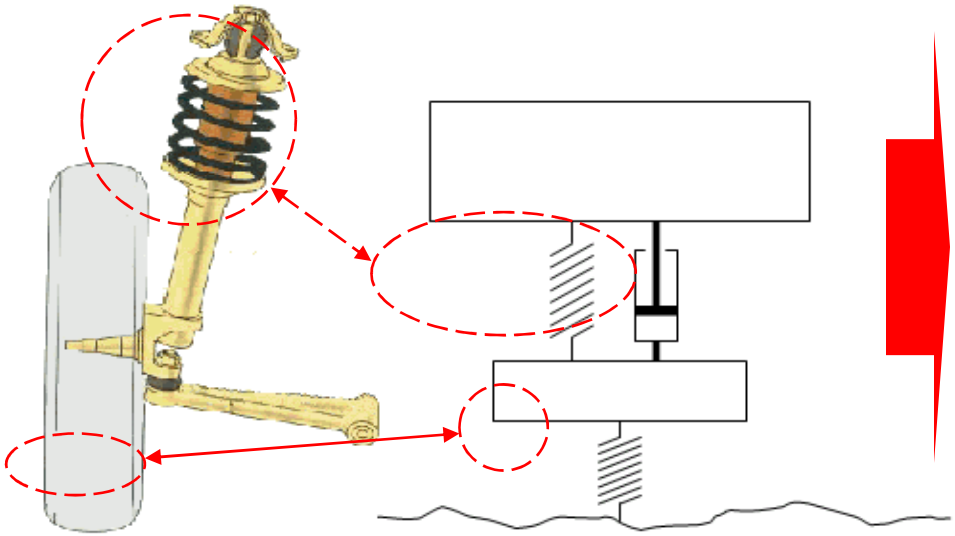
In term of velocity

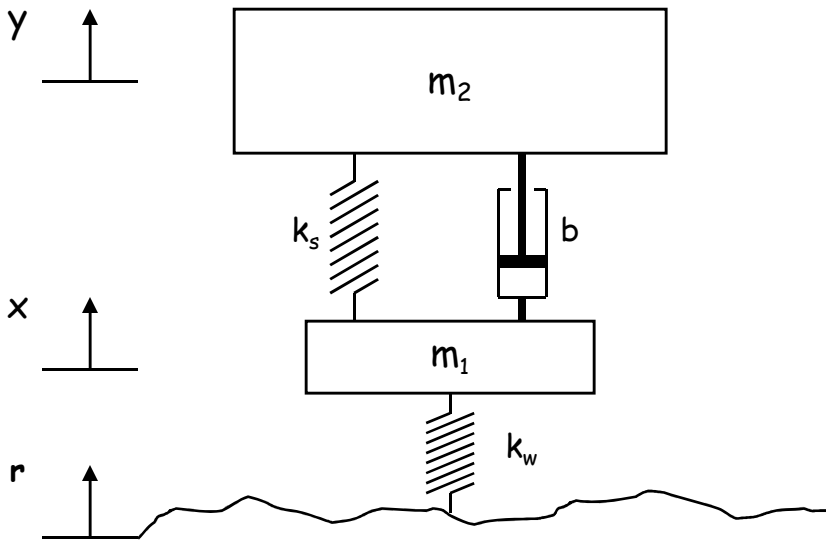
$$\dot{v} + \frac{b}{m}v = \frac{u}{m}$$

Dynamics of Mechanical Systems

Example 3

Obtain the mathematical model for the suspension system of the car!

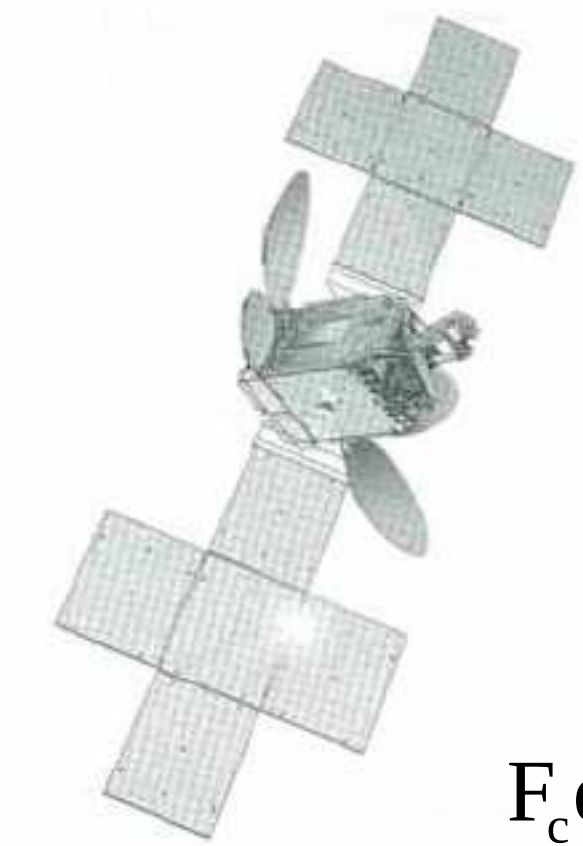




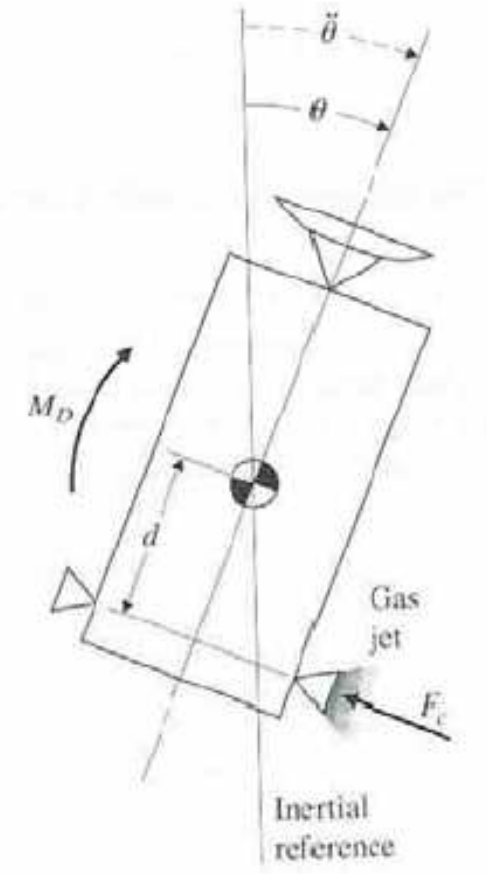
A quarter car model

Dynamics of Mechanical Systems

Example 4



FBD?



$$F_c d + M_D = I\ddot{\theta}$$

Electrical System

What is electrical system?

Electrical systems are concerned with the behavior of three fundamental quantities: charge, current and voltage.

Kirchoff's laws are used to model the electrical system:

- Kirchoff's Voltage Law (KVL)
- Kirchoff's Current Law (KCL).

Electrical System

- Kirchoff's Voltage Law (KVL)

  The sum of voltage around a closed loop or path is zero.

$$\sum_{i=1}^{N} v_i = 0$$

Electrical System

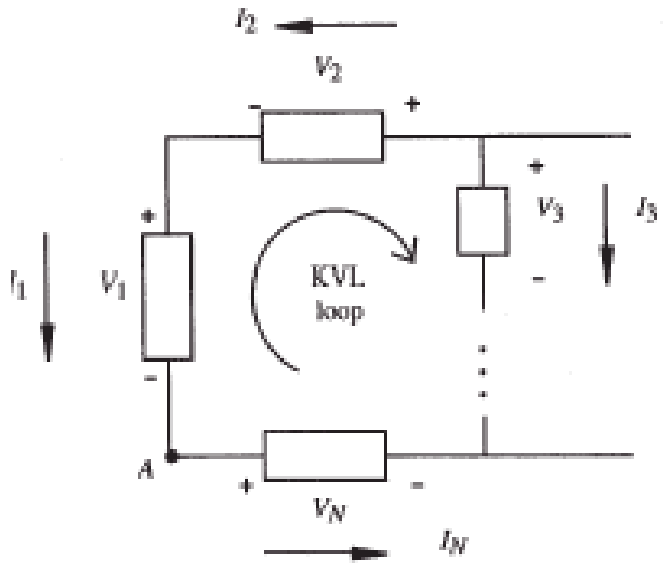
- Kirchoff's Voltage Law (KVL)

Approach:

- Assume the current direction for each element
- Assign the appropriate polarity for each element
- Assume the KVL loop either CW or CCW
- Start at any point and assign positive sign if the assumed current direction is the same with KVL loop and vice versa
- If a voltage result is negative, the voltage actually drops in the opposite direction.

Electrical Systems

- Kirchoff's Voltage Law (KVL)



?

$$-V_1 - V_2 + V_3 + \cdots + V_N = 0$$

Electrical Systems

- Kirchoff's Current Law (KCL)

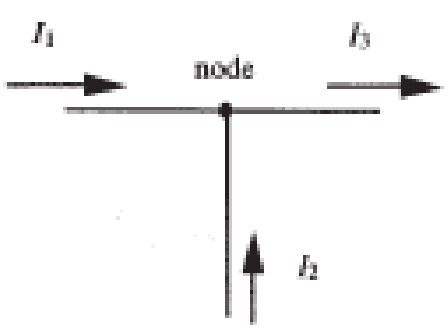
The sum of current flowing into a closed surface or node is zero.

$$\sum_{n=1}^{N} i_n = 0$$

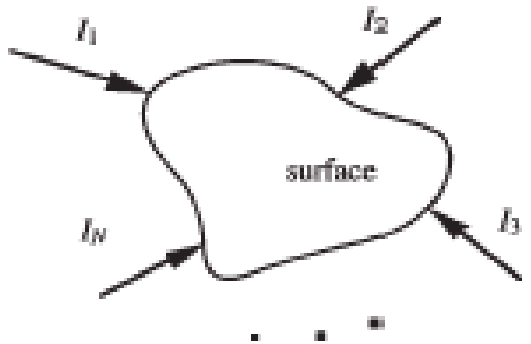
Electrical Systems

- Kirchoff's Current Law (KCL)

  Approach:
  - Assume the current directions arbitrary
  - If a current is leaving a node, assign negative sign
  - If the calculated current is negative sign, the current actually flows in the opposite direction.



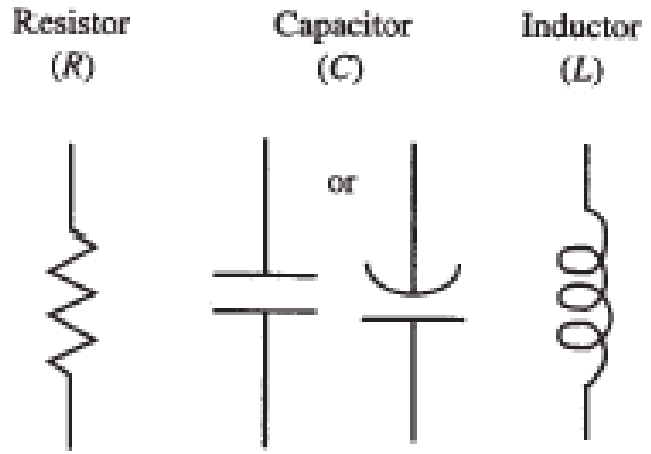
a) example KCL



b) general KCL

Electrical Systems

Basic Electrical Element of Electrical Systems



R: resistance in Ohm
C: capacitance in F
L: inductance in H

Voltage and current characteristics:

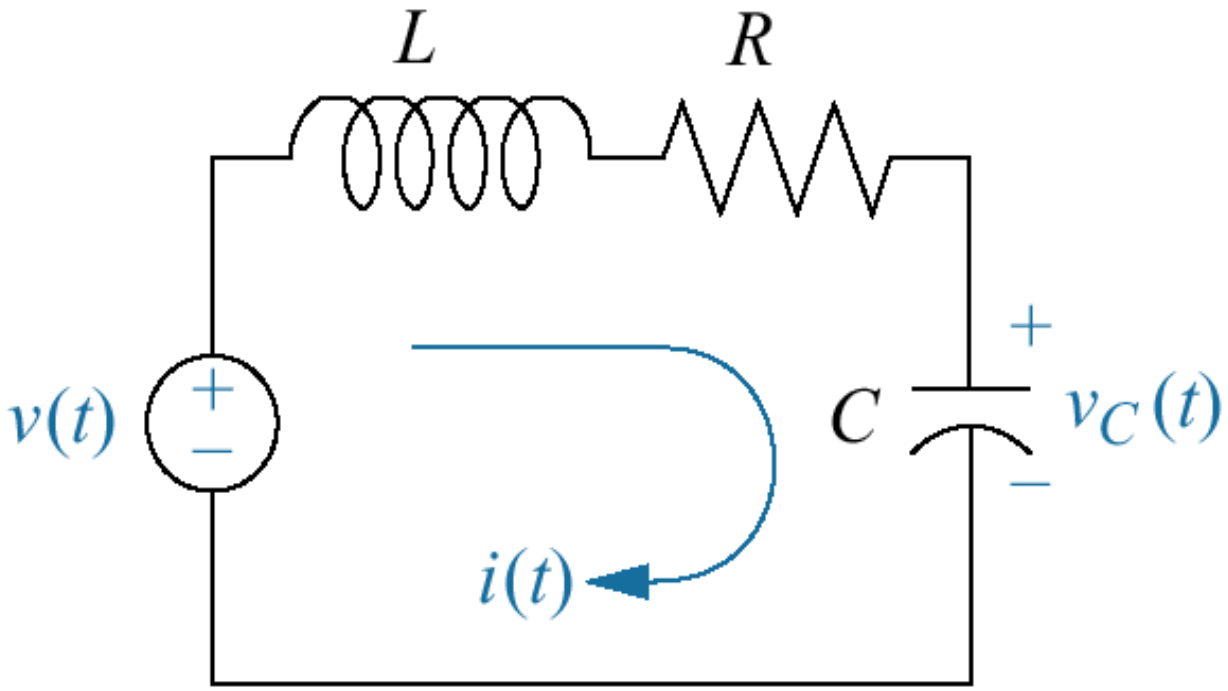
Resistor: $v = Ri$

Capacitor: $i = C\dfrac{dv}{dt}$

Inductor: $v = L\dfrac{di}{dt}$

Electrical Systems

Example:



Electrical Systems

What is electromechanical system?

Electromechanical systems are systems governed by both Newton's and Kirchoff Laws simultaneously.

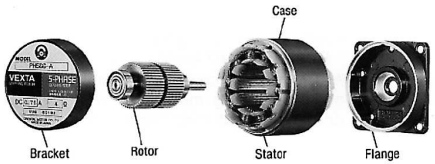
Example of electromechanical system:

- Motor: DC, AC, stepper motors, etc
- Generator
- Loudspeaker
- etc

Electromechanical Systems

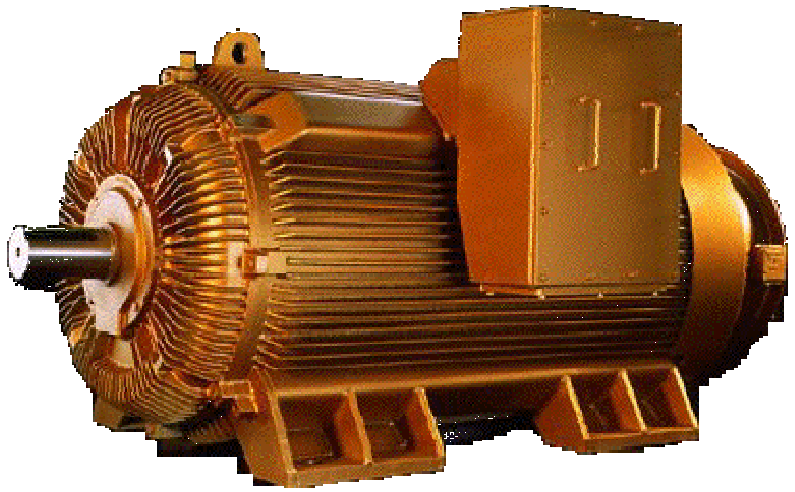


(*a*) AC induction and stepper motor



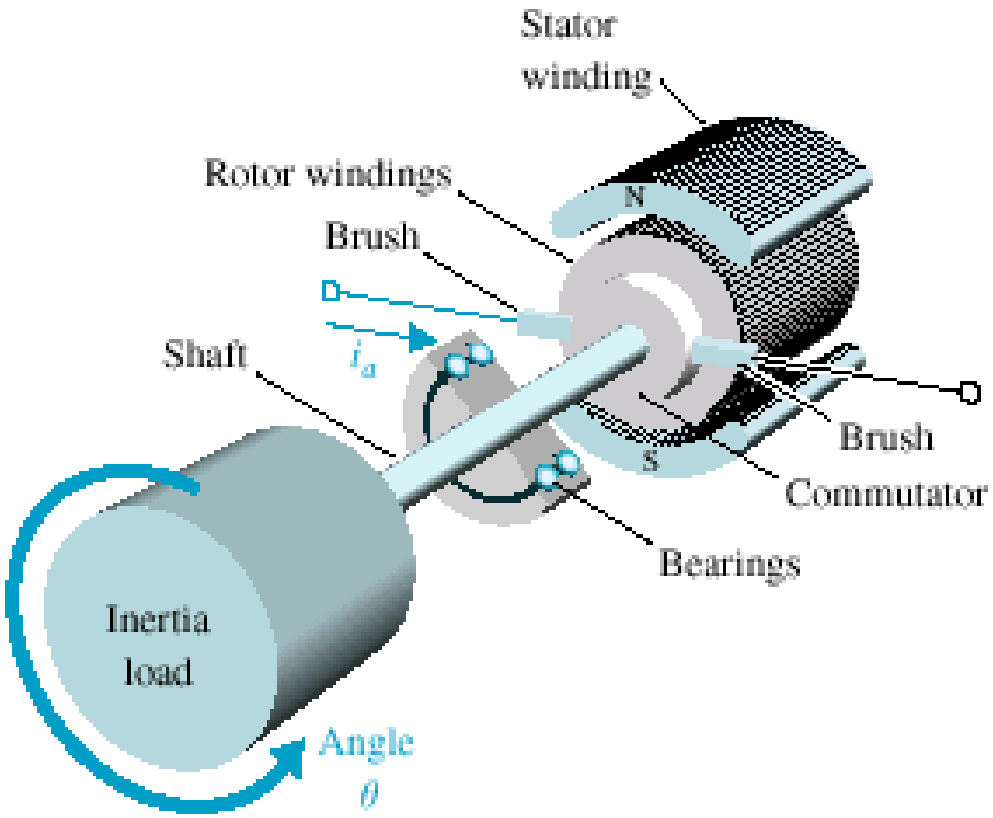
(*b*) exploded view of stepper motor with a permanent magnet rotor

Figure 10.6 Examples of commercial motors. *(Courtesy of Oriental Motor, Torrance, CA)*



DC Motor

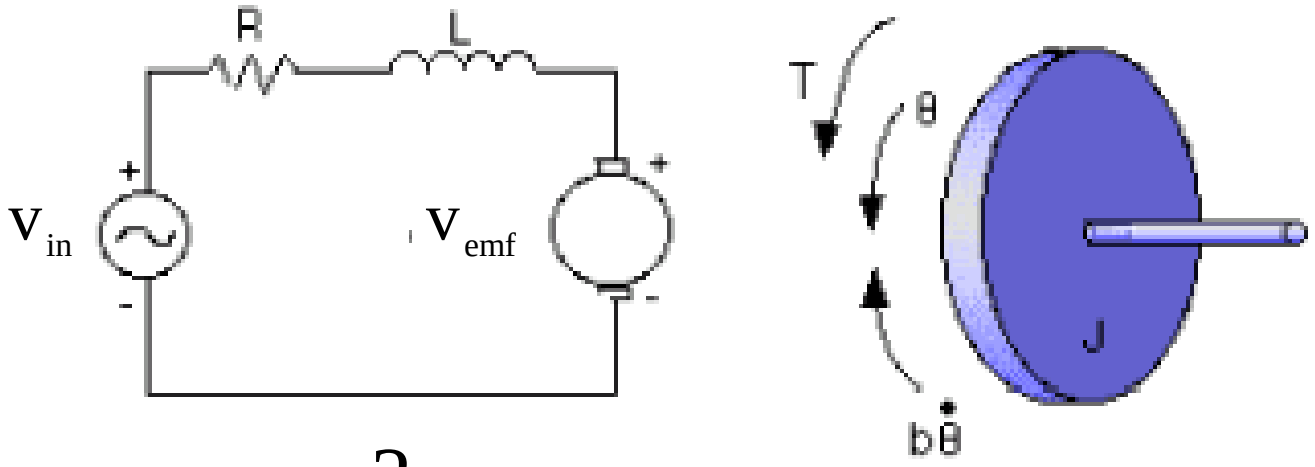
Electromechanical Systems



DC Motor Model?

Electromechanical Systems

Modeling of DC Motor



$$v_{in} = ?$$

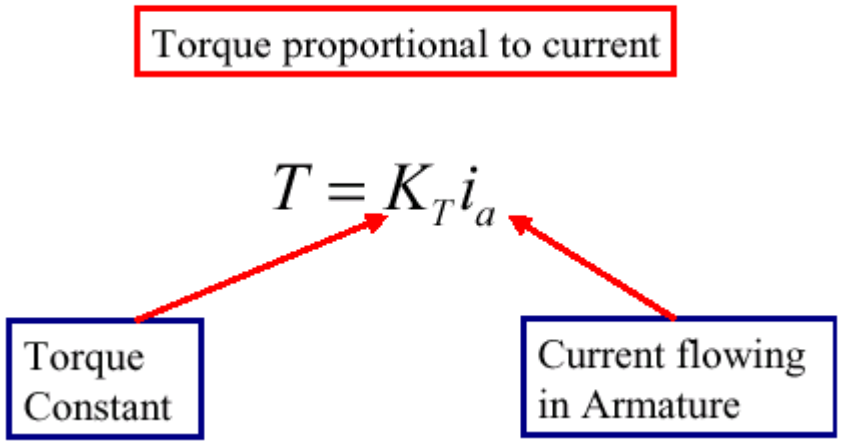
The back-emf is proportional to the speed of the motor ω in rad/s as

$$v_{emf} = K_e\dot{\theta} \quad \Rightarrow \quad v_{in} = L\frac{di}{dt} + Ri + K_e\dot{\theta}$$

Electromechanical Systems

Modeling of DC Motor

The torque generated by a DC motor is directly proportional to armature current



Electromechanical Systems

Modeling of DC Motor

The torque generated is used to drive the loads:

$$T = \left(J_m + J_L\right)\ddot{\theta} + b\dot{\theta}$$

Where:
J_m: motor inertia
J_L: attached load inertia
b : viscous damping

Summary

- Kirchoff′s law is used to obtain mathematical model of electrical systems.
- Newton′s and Kirchoff′s laws are used to obtain mathematical model of electromechanical systems.
- Challenge:
  Modeling of a given mechanical system!